

Day 28 - OSPF (pt 3)

Sunday, September 6, 2026 10:46 AM

OSPF Information

Loopback Interfaces

A loopback interface is a virtual interface in the router that's always in an **up/up** state (unless shutdown).

- It does not depend on a physical interface, therefore, it provides a consistent IP address that can be used to reach the router or identify it.

OSPF Network Types

OSPF has a few network types that are used for different scenarios and behave differently. The 3 main ones are:

1. **Broadcast**
 - Enabled by default on **Ethernet** and **FDDI** (*Fiber Distributed Data Interface*) interfaces.
2. **Point-to-point**
 - Enabled by default on **PPP** (*Point-to-Point Protocol*) and **HDLC** (*High-Level Data Link Control*).
3. **Non-broadcast**
 - Enabled by default on **Frame Relay** and **X.25** interfaces.

[How to change the network type on an interface?](#)

The OSPF network type can be configured on an interface with the `ip ospf network NETWORK-TYPE` command

Network Type Table Summary

Broadcast	Point-to-point	Non-broadcast
Default on Ethernet and FDDI	Default on PPP and HDLC	Default on Frame Relay and X.25
DR/BDR elected	No DR/BDR elected	No DR/BDR elected
Neighbors dynamically discovered	Neighbors dynamically discovered	Neighbors aren't dynamically discovered
Default timers: Hello - 10 seconds Dead - 40 seconds	Default timers: Hello - 10 seconds Dead - 40 seconds	Default timers: Hello - 30 seconds Dead - 120 seconds

Broadcast Network Type

The broadcast network type is enabled by default on **Ethernet** and **FDDI** (*Fiber Distributed Data Interface*) interfaces.

- Routers **dynamically discover** neighbors by sending and receiving **OSPF Hello** messages using the multicast address of 224.0.0.5
- A **DR** (**Designated Router**) and **BDR** (**Backup Designated Router**) must be elected on each subnet. (With there only being a **DR** if there are no **OSPF** neighbors)
 - Any other routers in the network which aren't **DRs** or **BDRs** will become **DROthers**.
 - Messages to the **DR/BDR** are sent to the multicast address of 224.0.0.6

Important DR/BDR and DROthers info

- Routers will only form a **FULL OSPF** adjacency with the **DR** and **BDR** of the segment. Therefore, routers only exchange **LSAs** with the **DR** and **BDR**.
- Meanwhile, the **DROthers** establish a **2-way** connection between each other, and do not share **LSAs** with each other. (*All routers will still have the same LSDB, but this reduces the amount of LSAs flooding the network.*)
- In short,
 - the **DR** and **BDR** will form a **FULL OSPF** adjacency with **ALL** routers in the subnet.
 - **DROthers** will form a **FULL OSPF** adjacency **only** with the **DR/BDR**

DR and BDR Election

There are two values that are checked when electing a DR and BDR, with the order of priority being from top to bottom

1. Highest **OSPF** interface **priority**

2. Highest **OSPF Router ID**

The first place is assigned the DR role for the subnet, the second place is assigned the BDR role. All other routers in the subnet will become DROthers.

- The **DR/BDR** election is '*non-preemptive*', meaning that once both the **DR** and **BDR** are selected, they will keep their role until the **OSPF** process is reset, or the interface is shutdown/fails, etc.
 - If the **OSPF** process is reset or the **DR/BDR** interface is down for whatever reason, the **BDR** will instantly assume the role of **DR** and a *new BDR* election will be held, even if there are currently interfaces with higher *priorities* or *Router IDs*.

OSPF Interface Priority

The default OSPF interface *priority* is **1** on all interfaces.

If the OSPF interface *priority* is set to **0**, the router cannot become a DR or BDR for the subnet.

The priority can be changed on a per-interface basis with a maximum value of **255** using the command:

ip ospf priority PRIORITY-VALUE

Point-to-Point Network Type

The Point-to-Point network type is enabled by default on **PPP** (*Point-to-Point Protocol*) and **HDLC** (*High-Level Data Link Control*).

- Routers *dynamically discover* neighbors by sending and receiving **OSPF Hello** messages using the multicast address of **224.0.0.5**
- There are no **DR/BDR** elections, as this connection type is used between two devices (aka **point-to-point**), so there would be no need to elect a **DR** or **BDR**.

Serial Interfaces

Serial interfaces are old interfaces that were used before the invention of Ethernet interfaces.

- It consists of two connections with one side functioning as the **DCE** (**Data Communications Equipment**) and the other functioning as the **DTE** (**Data Terminal Equipment**).
- The **DCE** side of the connection is the one responsible for the *clock rate* (*speed*) of the connection.
- The default *encapsulation* on a **serial** interface is **HDLC**. (*cHDLC for Cisco devices*)
 - The *encapsulation* type must match on both sides of the connection, or else, both interfaces will go *down* won't be able to communicate with each other.

Clock rate VS. Speed

Ethernet interfaces use the *speed MEGABITS-PER-SECOND* command to configure their interfaces' working speed.

Meanwhile, Serial interfaces use the *clock rate BITS-PER-SECOND* command

How to figure out which side is DCE and which is DTE?

You can check which side is DCE and DTE with the *show controllers serial INTERFACE-ID* command. Where *INTERFACE-ID* is the ID of the serial interface you're checking.

Broadcast	Point-to-point
Default on Ethernet , FDDI interfaces	Default on HDLC , PPP (serial) interfaces
DR/DBR elected	No DR/BDR
Neighbors dynamically discovered	Neighbors dynamically discovered
Default timers: Hello 10, Dead 40	Default timers: Hello 10, Dead 40

OSPF Neighbor Requirements

For OSPF routers to become neighbors, there are multiple requirements they should meet. The most important ones are:

- OSPF** process must not be shutdown

1. **OSPF** process must not be shutdown
2. Interfaces must be in the same subnet
3. **Area** number must match
4. **OSPF Router IDs** must be unique
5. **Hello** and **Dead** timers must match
6. Authentication settings must match
7. IP MTU settings must match
8. **OSPF** Network Type must match
[IP MTU and Network Type disclaimer](#)

For **#7** (IP MTU settings must match) and **#8** (OSPF Network Type must match), routers can become OSPF neighbors with each other, however, they will not operate properly.

- So, if you check the **OSPF show** commands and everything looks fine but the system isn't functioning correctly, make sure to check on **#7** and **#8**

OSPF LSA Types

The OSPF LSDB contains a list of LSAs, and, there are 11 different types of LSAs that can be contained within that database. The most important three for the CCNA are:

1. **Type 1 (Router LSA)**
 - Every **OSPF** router generates this **LSA**.
 - It identifies the router using the **Router ID**.
 - It lists the networks attached to the router's **OSPF**-activated interfaces.
2. **Type 2 (Network LSA)**
 - Generated by the **DR** of each **multi-access** network (like the **broadcast** network type)
 - Lists the routers which are attached to the **multi-access** network.
3. **Type 5 (AS-External LSA)**
 - Generated by the **ASBRs** to describe routes to destinations outside of the **AS** (**OSPF** domain)